

Introduction to Optimization Project 2

Jun Moon

junmoon@hanyang.ac.kr

Project 2

- Group Project
- Submission with zip file (압축해서 제출해주세요)
 - Programming files (py-files)
 - Solution PDF
 - PPT Slides
 - Presentation MP4 (25 mins)

Submission

- 1. PPT Slides of your presentation**
- 2. The explanation of your solution in English (PDF Format)**
- 3. Python Codes**
- 4. Presentation MP4 (25 mins)**

Note

- **You can use Python to solve the problem**
- **You have to construct the algorithm by yourself**
- **You are not able to use any Python optimization toolbox**
 - **You need to implement the numerical optimization algorithms**

Required Python Libraries

- Numpy
- Matplotlib
- Scipy
- Pandas
- etc

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- Numpy
- Matplotlib
- Scipy

Problem 1

- In class, we have obtained the dynamic programming algorithm for the Knapsack problem to find the optimal cost
- However, there is no way to find the optimal Knapsack strategy
- Problem
 - You have to implement the dynamic programming algorithm to find the optimal cost and the optimal Knapsack strategy.

Problem 2

- Study Branch and Bound Algorithm for Knapsack problem
- You can use the following textbook
 - <https://link.springer.com/book/10.1007/978-3-540-24777-7>

Branch and Bound

- Land and Doig (1960) for discrete programming

ECONOMETRICA

VOLUME 28

July, 1960

NUMBER 3

AN AUTOMATIC METHOD OF SOLVING DISCRETE PROGRAMMING PROBLEMS

BY A. H. LAND AND A. G. DOIG

[www.jstor.org/stable](http://www.jstor.org/stable/2353111) ▼ [이 페이지 번역하기](#)

[An Automatic Method of Solving Discrete Programming ... - jstor](#)

AH Land 저술 - 1960 - 3531회 인용 - [관련 학술자료](#)

AN AUTOMATIC METHOD OF SOLVING DISCRETE, PROGRAMMING PROBLEMS. By A. H. LAND AND A. G. DOIG. In the classical linear programming ...

Branch and Bound (Knapsack)

- Kolesar (1967)

www0.gsb.columbia.edu › kolesar_branch_bound ▼ PDF [이 페이지 번역하기](#)

A Branch and Bound Algorithm for the Knapsack Problem

PJ KOLESAR 저술 - 1967 - 270회 인용 - 관련 학술자료

PETER J. KOLESAR. Columbia University. A branch and bound algorithm for solution of the "knapsack problem," $\max \sum v_i x_i$ where $\sum w_i x_i < W$ and $x_i = 0, 1, \dots$

MANAGEMENT SCIENCE
Vol. 13, No. 9, May, 1967
Printed in U.S.A.

A BRANCH AND BOUND ALGORITHM FOR THE KNAPSACK PROBLEM*†

PETER J. KOLESAR

Columbia University

A branch and bound algorithm for solution of the "knapsack problem," $\max \sum v_i x_i$ where $\sum w_i x_i \leq W$ and $x_i = 0, 1$, is presented which can obtain either optimal or approximate solutions. Some characteristics of the algorithm are discussed and computational experience is presented. Problems involving 50 items from which approximately 25 items were loaded were solved in an average of 0.07 minutes each by a coded version of this algorithm for the IBM 7094 computer.

- **function output**
 - **optimal cost of the Knapsack problem**
 - **optimal Knapsack strategy (optimal x^*)**
 - **Cost comparison**
 - **Greedy algorithm**
 - **Optimal Knapsack cost (by Dynamic programming and Branch and bound)**
 - **LP relaxation and its round down value**
 - **you can use the following LP command**
(<https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.linprog.html>)

Must show the result of your dynamic programming and branch and bound algorithms

- Generate the Knapsack data by your self
- You can choose any size of the vector and matrices
- Cost comparison
 - Optimal Knapsack cost (by dynamic programming and branch and bound)
 - LP relaxation
 - Greedy algorithm

You need to verify your optimal solution

- You can use “intlinprog” in MATLAB, which is the integer programming solver in MATLAB (help intlinprog)

- Your algorithms must be independent of the size of the Knapsack problem
- Your algorithms must be independent of the structure of the Knapsack problem
 - You need to design and check the assumptions of the Knapsack problem
- You can choose any size of the vector and matrices
- You have to compare your optimal solution with the solution obtained by other software package such as “intlinprog” in MATLAB